Imperial College London

October 29, 2026

Department of Computing

Human-Robot Interaction – Dr Nicole Salomons

Assessed Coursework 2

The Coursework should be submitted on Scientia by November 5th 2026, 7pm, and consist in:
- Python files of your code with instructions on how to run it, a short report (1 page maximum) briefly explaining how it works, and an mp4 video of you interacting with the simulated reachy Mini. Please keep the video to less than 30s long.
The Scientia submission will open on October 30th. Please ensure that you are familiar with it before the deadline. Unfortunately, emailed or printed submissions cannot be accepted. Plan to spend around four to seven hours of your time on this coursework, of which two will be spent in class.

You are encouraged to discuss the general Coursework questions with other students, but your answers should be yours. We also encourage you to test your joint attention system on several colleagues and users to ensure that it generalizes to other users. You are welcome to use AI tools for programming this assignment, however you must understand how the code works.

Marks are shown at the bottom of each section. Note, that the **marks are only indicative**. If you have questions about the Coursework please make use of EdStem, but note that the module leader and Teaching Assistants cannot provide you with answers that directly solve the Coursework.

1 Point, Look, and Name (100 points)

Develop a Reachy Mini application in which a user points at an object visible in a webcam frame. The robot must determine which object is being indicated, turn its head toward it in the MuJoCo simulation, and name the object.

1.1 Objectives

The completed system should:

- detect a pointing gesture;
- recognise common objects in the camera frame;
- determine which object the user is pointing toward;
- visibly turn Reachy's head toward the selected object; and
- display and speak the object's name.

1.2 Technical Requirements

You must use:

- Python and the Reachy Mini SDK;
- MediaPipe Hand Landmarker for detecting the pointing hand;
- A suitable model for object detection; and
- the full MuJoCo simulation so that Reachy's head movement is visible.

The target object must be selected using your own rule-based (deterministic) method based on the hand landmarks and detected object locations. For example, you may construct a pointing ray and compare it with the detected bounding boxes.

You may choose your own thresholds, stability rules, interface, software architecture, and head-movement mapping, provided that you explain your decisions.

1.3 Expected Behaviour

Your demonstration should include:

- at least two objects visible at the same time;
- correct selection when the user points at either object;
- visible Reachy head movement in the simulation;
- an annotated camera preview showing the pointing direction and detected objects;
- (bonus) sensible handling of unclear or ambiguous pointing;

We are not looking for perfect object-recognition accuracy. A system that refuses to guess when the pointing direction is unclear is preferable to one that frequently selects the wrong object.

1.4 Submission

Submit:

- the complete source code;
- a README explaining how to run the application;
- a short demonstration video; and
- a very brief report explaining your approach.

1.5 Grading

- Pointing-gesture detection (20 points)
- Object-detection integration (20 points)
- Rule-based (deterministic) target-selection method (20 points)
- Reachy head movement and object naming (20 points)
- Code quality, report, and clarity (20 points)